

HOMEWORK 5 SOLUTIONS

SECTION 2.6

② a

Can we write $(1, 3, -2, -3)$ as a linear combination of $(1, 1, 0, -1)$ and $(0, -1, 1, 1)$?

$$\begin{bmatrix} 1 \\ 3 \\ -2 \\ -3 \end{bmatrix} = a \begin{bmatrix} 1 \\ 1 \\ 0 \\ -1 \end{bmatrix} + b \begin{bmatrix} 0 \\ -1 \\ 1 \\ 1 \end{bmatrix} \Leftrightarrow \begin{cases} 1 = a \\ 3 = a - b \\ -2 = b \\ -3 = -a + b \end{cases} \quad \text{This system has a (unique) solution } a=1, b=-2.$$

Thus,

$$\begin{bmatrix} 1 \\ 3 \\ -2 \\ -3 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 0 \\ -1 \end{bmatrix} - 2 \begin{bmatrix} 0 \\ -1 \\ 1 \\ 1 \end{bmatrix}, \text{ so } T \begin{bmatrix} 1 \\ 3 \\ -2 \\ -3 \end{bmatrix} = T \begin{bmatrix} 1 \\ 1 \\ 0 \\ -1 \end{bmatrix} - 2 T \begin{bmatrix} 0 \\ -1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 2 \\ 3 \\ -1 \end{bmatrix} - 2 \begin{bmatrix} 5 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} -8 \\ 3 \\ -3 \end{bmatrix}$$

↑
T linear

b Same idea:

$$\begin{bmatrix} 5 \\ -1 \\ 2 \\ -4 \end{bmatrix} = a \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} + b \begin{bmatrix} -1 \\ 1 \\ 0 \\ 2 \end{bmatrix} \Leftrightarrow \begin{cases} 5 = a - b \\ -1 = a + b \\ 2 = a \\ -4 = a + 2b \end{cases} \quad \text{This system has a unique solution } a=2, b=-3.$$

Therefore,

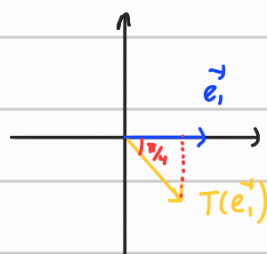
$$\begin{bmatrix} 5 \\ -1 \\ 2 \\ -4 \end{bmatrix} = 2 \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} - 3 \begin{bmatrix} -1 \\ 1 \\ 0 \\ 2 \end{bmatrix} \Rightarrow T \begin{bmatrix} 5 \\ -1 \\ 2 \\ -4 \end{bmatrix} = 2 T \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} - 3 T \begin{bmatrix} -1 \\ 1 \\ 0 \\ 2 \end{bmatrix} = 2 \begin{bmatrix} 5 \\ 1 \\ -3 \end{bmatrix} - 3 \begin{bmatrix} 2 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 4 \\ 2 \\ -9 \end{bmatrix}$$

③ b

If $T(\vec{x}) = -\vec{x}$ for each $\vec{x}$ in $\mathbb{R}^2$, then $T(\vec{e}_1) = T \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} -1 \\ 0 \end{bmatrix}$ and $T(\vec{e}_2) = T \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ -1 \end{bmatrix}$, so the matrix of T is

$$A = [T(\vec{e}_1) \quad T(\vec{e}_2)] = \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix}$$

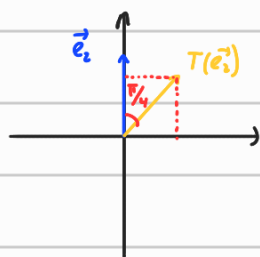
c Let's find $T(\vec{e}_1)$ and $T(\vec{e}_2)$:



From the picture, $T(\vec{e}_1) = \begin{bmatrix} \cos \pi/4 \\ -\sin \pi/4 \end{bmatrix} = \begin{bmatrix} 1/\sqrt{2} \\ -1/\sqrt{2} \end{bmatrix}$

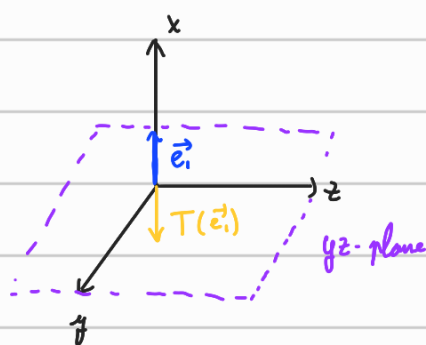
We conclude that the matrix of T is

$$A = [T(\vec{e}_1) \quad T(\vec{e}_2)] = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix}$$

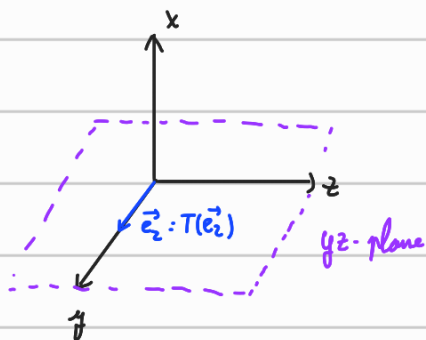


Similarly, we get $T(\vec{e}_2) = \begin{bmatrix} \cos \pi/4 \\ \sin \pi/4 \end{bmatrix} = \begin{bmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \end{bmatrix}$

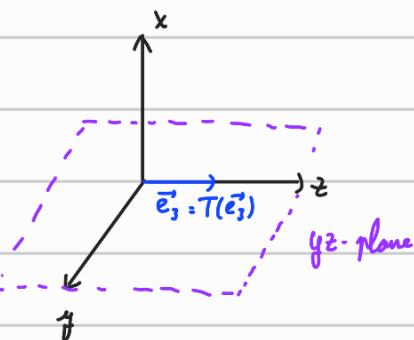
46 Let's find $T(\vec{e}_1)$, $T(\vec{e}_2)$ and $T(\vec{e}_3)$:



The reflection of $\vec{e}_1$ along the yz -plane is $T(\vec{e}_1) = -\vec{e}_1 = \begin{bmatrix} -1 \\ 0 \\ 0 \end{bmatrix}$



The reflection of $\vec{e}_2$ along the yz -plane is $T(\vec{e}_2) = \vec{e}_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$ because $\vec{e}_2$ is contained in the yz -plane.



The same reasoning as with $\vec{e}_2$ gives $T(\vec{e}_3) = \vec{e}_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$.

Therefore, the matrix of T is $A = [T(\vec{e}_1) \ T(\vec{e}_2) \ T(\vec{e}_3)] = \begin{bmatrix} -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

⑦ a In order for T to be linear, it has to satisfy $T(\vec{x} + \vec{y}) = T(\vec{x}) + T(\vec{y})$ and $T(c\vec{x}) = cT(\vec{x})$ for all $\vec{x}, \vec{y}$ in $\mathbb{R}^3$ and c in $\mathbb{R}$. We show that, for instance, T does not satisfy the second equation in general.

Take $\vec{x} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$, $c = 2$. Then

$$T(c\vec{x}) = T\left(\begin{bmatrix} 2 \\ 2 \end{bmatrix}\right) = \begin{bmatrix} 4 \\ 0 \end{bmatrix}$$

$$cT(\vec{x}) = 2T\left(\begin{bmatrix} 1 \\ 1 \end{bmatrix}\right) = 2\begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 2 \\ 0 \end{bmatrix}.$$

So $T(c\vec{x}) \neq cT(\vec{x})$, meaning that T is not linear.

b We can argue very similarly:

Take $\vec{x} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$ and $c = 2$. Then

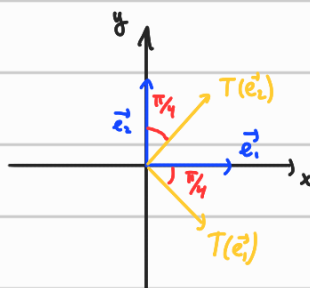
$$T(c\vec{x}) = T\left(\begin{bmatrix} 0 \\ 2 \end{bmatrix}\right) = \begin{bmatrix} 0 \\ 4 \end{bmatrix}$$

$$cT(\vec{x}) = 2T\left(\begin{bmatrix} 0 \\ 1 \end{bmatrix}\right) = 2\begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 2 \end{bmatrix}$$

Because $T(c\vec{x}) \neq cT(\vec{x})$, we conclude that T is not linear.

8b What does T do to the standard basis vectors? We have $T(\vec{e}_1) = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ -1 \end{bmatrix}$ and $T(\vec{e}_2) = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix}$.

Graphically, we have



The vector $T(\vec{e}_1)$ is the result of rotating $\vec{e}_1$ clockwise by an angle of $\pi/4$, and similarly with $T(\vec{e}_2)$. Therefore, T is the clockwise rotation through an angle of $\pi/4$ radians (alternatively, the counterclockwise rotation through $2\pi - \frac{\pi}{4} = \frac{7\pi}{4}$ radians).

SECTION 3.1

① e

$$\begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix} = \cos^2 \theta + \sin^2 \theta = \boxed{1}$$

f

$$\begin{vmatrix} 2 & 0 & -3 \\ 1 & 2 & 5 \\ 0 & 3 & 0 \end{vmatrix} = (-3) \begin{vmatrix} 2 & -3 \\ 1 & 5 \end{vmatrix} = (-3)(10+3) = \boxed{-39}$$

Expand along
3rd row

h

$$\begin{vmatrix} 0 & a & 0 \\ b & c & d \\ 0 & e & 0 \end{vmatrix} = -a \begin{vmatrix} b & d \\ 0 & e \end{vmatrix} = \boxed{0}$$

Expand along
2nd row Row of
 zeros

k

$$\begin{vmatrix} 0 & 1 & -1 & 0 \\ 3 & 0 & 0 & 2 \\ 0 & 1 & 2 & 1 \\ 5 & 0 & 0 & 7 \end{vmatrix} \xrightarrow{C_3+C_2} \begin{vmatrix} 0 & 1 & 0 & 0 \\ 3 & 0 & 0 & 2 \\ 0 & 1 & 3 & 1 \\ 5 & 0 & 0 & 7 \end{vmatrix} \xrightarrow{\text{Expand along 1st row}} - \begin{vmatrix} 3 & 0 & 2 \\ 0 & 3 & 1 \\ 5 & 0 & 7 \end{vmatrix} \xrightarrow{\text{Expand along 2nd row}} -3 \begin{vmatrix} 3 & 2 \\ 5 & 7 \end{vmatrix} = -3(21-10) = \boxed{33}$$

Expand along
2nd row Expand along
1st row 2nd row

n

$$\begin{vmatrix} 4 & -1 & 3 & -1 \\ 3 & 1 & 0 & 2 \\ 0 & 1 & 2 & 2 \\ 1 & 2 & -1 & 1 \end{vmatrix} \xrightarrow{\substack{R_2-3R_4 \\ R_1-4R_4}} \begin{vmatrix} 0 & -9 & 7 & -5 \\ 0 & -5 & 3 & -1 \\ 0 & 1 & 2 & 2 \\ 1 & 2 & -1 & 1 \end{vmatrix} \xrightarrow{\text{Expand along 1st column}} - \begin{vmatrix} -9 & 7 & -5 \\ -5 & 3 & -1 \\ 1 & 2 & 2 \end{vmatrix} \xrightarrow{\substack{C_2-2C_1 \\ C_3-3C_1}} \begin{vmatrix} -9 & 25 & 13 \\ -5 & 13 & 9 \\ 1 & 0 & 0 \end{vmatrix} \xrightarrow{\text{Expand along 3rd row}} - \begin{vmatrix} -9 & 25 & 13 \\ -5 & 13 & 9 \end{vmatrix} = -(225-161) = \boxed{-56}$$

Expand along
1st column Expand along
3rd row

② Suppose $A = [a_{ij}]$ is an $n \times n$ matrix that has a row of zeros, say, the i -th row. This means that $a_{i1} = a_{i2} = \dots = a_{in} = 0$.

If we compute $\det(A)$ by taking the cofactor expansion along the i -th row, we get

$$\det(A) = a_{i1}C_{i1}(A) + a_{i2}C_{i2}(A) + \dots + a_{in}C_{in}(A) = 0 \cdot C_{i1}(A) + 0 \cdot C_{i2}(A) + \dots + 0 \cdot C_{in}(A) = 0$$

If A has a column of zeros, one can show that $\det(A) = 0$ in a similar fashion. Another option is to note that A^T has a row of zeros, so $\det(A) = \det(A^T) = 0$ from the above argument.

⑥ a

$$\begin{vmatrix} a & b & c \\ a+1 & b+1 & c+1 \\ a-1 & b-1 & c-1 \end{vmatrix} \xrightarrow{\substack{R_1 \leftarrow R_1 \\ R_2 \leftarrow R_2 \\ R_3 \leftarrow R_3}} \begin{vmatrix} a & b & c \\ 1 & 1 & 1 \\ -1 & -1 & -1 \end{vmatrix} \xrightarrow{\text{Two equal rows}} \begin{vmatrix} a & b & c \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{vmatrix} = 0$$

b

$$\begin{vmatrix} a & b & c \\ a+b & 2b & c+b \\ 2 & 2 & 2 \end{vmatrix} \xrightarrow{R_2 \leftarrow R_2} \begin{vmatrix} a & b & c \\ a & b & c \\ 2 & 2 & 2 \end{vmatrix} \xrightarrow{\text{Two equal rows}} 0$$

⑦ a

$$\begin{vmatrix} -x & -y & -z \\ 3p+a & 3q+b & 3r+c \\ 2p & 2q & 2r \end{vmatrix} \xrightarrow{R_1 \leftarrow R_1} \begin{vmatrix} x & y & z \\ 3p+a & 3q+b & 3r+c \\ p & q & r \end{vmatrix} \xrightarrow{R_1 \leftarrow R_1} \begin{vmatrix} x & y & z \\ a & b & c \\ p & q & r \end{vmatrix} \xrightarrow{R_1 \leftrightarrow R_2} \begin{vmatrix} a & b & c \\ x & y & z \\ p & q & r \end{vmatrix} \xrightarrow{R_1 \leftrightarrow R_2} \begin{vmatrix} x & y & z \\ a & b & c \\ p & q & r \end{vmatrix} \xrightarrow{R_1 \leftrightarrow R_2} \begin{vmatrix} a & b & c \\ p & q & r \\ x & y & z \end{vmatrix} \xrightarrow{R_1 \leftrightarrow R_2} \begin{vmatrix} a & b & c \\ p & q & r \\ x & y & z \end{vmatrix} \xrightarrow{R_1 \leftrightarrow R_2} \begin{vmatrix} a & b & c \\ p & q & r \\ x & y & z \end{vmatrix} = (-2) \cdot (-1) = 2$$

b

$$\begin{vmatrix} -2a & -2b & -2c \\ 2p+x & 2q+y & 2r+z \\ 3x & 3y & 3z \end{vmatrix} \xrightarrow{R_1 \leftarrow R_1} \begin{vmatrix} a & b & c \\ 2p+x & 2q+y & 2r+z \\ x & y & z \end{vmatrix} \xrightarrow{R_2 \leftarrow R_2} \begin{vmatrix} a & b & c \\ 2p & 2q & 2r \\ x & y & z \end{vmatrix} \xrightarrow{R_2 \leftarrow R_2} \begin{vmatrix} a & b & c \\ p & q & r \\ x & y & z \end{vmatrix} \xrightarrow{R_2 \leftarrow R_2} \begin{vmatrix} a & b & c \\ p & q & r \\ x & y & z \end{vmatrix} = -6 \cdot 2 \cdot (-1) = 12$$

⑨ a

False: If $A = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$ and $B = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$, then $\det(A+B) = \begin{vmatrix} 1 & 0 \\ 0 & 1 \end{vmatrix} = 1$ and $\det(A) + \det(B) = 0 + 0 = 0$.

c

True (Covered in 3.2). A direct proof is as follows:

If $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ is any 2×2 matrix, then $A^T = \begin{bmatrix} a & c \\ b & d \end{bmatrix}$ and $\det(A) = ad - bc = \det(A^T)$

e

TRUE: $\det(7A) = 7^2 \det(A) = 49 \det(A)$

g

FALSE: If $A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$, then $\det(-A) = \begin{vmatrix} -1 & 0 \\ 0 & -1 \end{vmatrix} = 1$ and $-\det(A) = -1$.